

Abstract Algebra - Homework 2

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Problem 1

(1)

$$ij = k, \quad ij = -ji \implies -ji = k \implies ji = -k$$

Hence, $ij \neq ji$, so multiplication is not commutative and $\mathbb{H}$ is a non-commutative ring.

- (2) **(a)** Let $x, y \in \mathbb{H}$. One can verify by direct computation that conjugation reverses the order of multiplication, $\overline{xy} = \bar{y}\bar{x}$. Note that for any $h = a + bi + cj + dk \in \mathbb{H}$, we have

$$N(h) = h\bar{h} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R},$$

so $N(h)$ lies in the center of $\mathbb{H}$ and commutes with all elements. Using this, we compute:

$$\begin{aligned} N(xy) &= (xy)\overline{(xy)} \\ &= (xy)(\bar{y}\bar{x}) \\ &= x(y\bar{y})\bar{x} \\ &= xN(y)\bar{x} \\ &= N(y)x\bar{x} \\ &= N(y)N(x) = N(x)N(y). \end{aligned}$$

- (b)** Since $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2$, it follows that

$$N(x) = 0 \iff a = b = c = d = 0 \iff x = 0.$$

Thus, $N(x) = 0$ if and only if $x = 0$.

- (3) Let $x \in \mathbb{H}$ be nonzero. From previous section, we have $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2 > 0$. Define the inverse of x by

$$x^{-1} := \frac{\bar{x}}{N(x)}.$$

Then:

$$xx^{-1} = x \cdot \frac{\bar{x}}{N(x)} = \frac{x\bar{x}}{N(x)} = \frac{N(x)}{N(x)} = 1, \quad x^{-1}x = \frac{\bar{x}}{N(x)} \cdot x = \frac{\bar{x}x}{N(x)} = \frac{N(x)}{N(x)} = 1.$$

Hence, every nonzero element of $\mathbb{H}$ is invertible.

Suppose $x, y \in \mathbb{H}$ with $x \neq 0$ and $xy = 0$. Then, $x^{-1}(xy) = (x^{-1}x)y = 1 \cdot y = y = 0$. Thus, $y = 0$, and similarly one shows that $x = 0$ if $y \neq 0$ and $xy = 0$. Therefore, $\mathbb{H}$ has no zero-divisors, and every nonzero element is invertible.

- (4) *Note: The problem appears to contain a typo, listing $\mathbb{Z} \cdot k$ twice and omitting $\mathbb{Z} \cdot 1$. We interpret H as the subring of $\mathbb{H}$ generated by $1, i, j, k$, and $w = \frac{1}{2}(1 + i + j + k)$.*

Define $H := \mathbb{Z} \cdot 1 + \mathbb{Z} \cdot i + \mathbb{Z} \cdot j + \mathbb{Z} \cdot k + \mathbb{Z} \cdot w$. Then $H \subset \mathbb{H}$, and we verify that H is a subring. First, $0 = 0 \cdot 1 \in H$, and $1 \in H$ by definition. Since the sum or difference of two integer combinations of $1, i, j, k, w$ is again such a combination, H is closed under addition and additive inverses.

For closure under multiplication, we note that $1, i, j, k \in \mathbb{H}$ and their products lie in the $\mathbb{Z}$ -span of $1, i, j, k$. Also, since $w = \frac{1}{2}(1 + i + j + k)$, and $\mathbb{H}$ is closed under multiplication, products like w^2, iw, jw , and kw are all in the $\mathbb{Z}$ -span of $1, i, j, k, w$, so they lie in H . Thus, H is closed under multiplication.

Therefore, H is a subring of $\mathbb{H}$. As $\mathbb{H}$ has no zero-divisors and $H \subset \mathbb{H}$, it follows that H has no zero-divisors.